

3.5.1 Sequence

سیکوئنس (Sequence) کا مطلب ہے کہ ہدایات بالکل اسی ترتیب سے ایک کے بعد ایک اوپر سے نیچے چلتی ہیں جس ترتیب سے لکھی گئی ہیں۔

CS XI

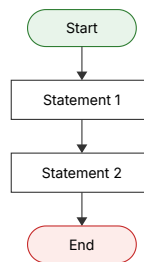
Chapter 3: Programming Fundamentals

Lecture 14 of 24

3.5.1 Sequence

LEARN IT

Sequence means statements are executed **one after another, strictly from top to bottom**, in the exact order they are written. This is the **default behaviour** of Python. No statement is skipped or repeated.



Sequence: control flows straight down from one statement to the next.

WORKED EXAMPLE

Input → Process → Output in sequence.

```
print("Enter two numbers")
a = int(input("Enter first number: ")) # Step 1: Input
b = int(input("Enter second number: ")) # Step 2: Input
total = a + b # Step 3: Process
print("Sum =", total) # Step 4: Output
```

Sample run (user types 7 and 5):

OUTPUT

```
Enter two numbers
Enter first number: 7
Enter second number: 5
Sum = 12
```

Each line runs exactly once, in order. If you moved line 4 (`total = a + b`) above the input lines, Python would stop with a `NameError` because `a` and `b` would not exist yet.

REMEMBER IT

Sequence is like reading a page: line 1, then line 2, then line 3. Most programs follow the pattern **Input → Process → Output**.

COMMON MISTAKE

Order matters. A variable must be given a value **before** it is used. Using it first causes `NameError: name '...' is not defined`.

اردو میں سمجھیں

سیکوئنس (Sequence) کا مطلب ہے کہ ہدایات بالکل اسی ترتیب سے ایک کے بعد ایک اوپر سے نیچے چلتی ہیں جس ترتیب سے لکھی گئی ہیں۔ یہ پائتھن کا عام طریقہ ہے۔ کسی ویری ایبل کو استعمال کرنے سے پہلے اس میں ویلیو رکھنا ضروری ہے۔

CHECK YOURSELF

1. Predict the output:

```
x = 4
y = x * 2
x = 10
print(x, y)
```

2. What is the default control structure in Python?

Answers: 1. 10 8 — `y` was calculated when `x` was still 4; changing `x` later does not change `y`. 2. Sequence.

Next lecture: 3.5.2 Selection Statements: if and if-else